StockSense

System Requirements Specification (SRS)

Prepared for: Codexus Technologies
Author: Kavindu Hasaranga (Electronics & IoT Trainee)
Date: 15th October 2026

Executive Summary: StockSense is an edge-connected, AI-driven smart inventory monitoring system. This SRS specifies the functional, performance, cost, environmental, and power requirements for designing, manufacturing, and validating commercial smart-shelf nodes for tropical and resource-constrained enterprise markets.

1. Functional Requirements

1.1 Sensing, Identification & Signal Processing

The system must accurately measure mass, isolate vibration, identify items, and monitor storage conditions in real time.

ID	Description	Target Mechanism	Priority
FR-SENS-01	Continuous weight measurement via strain-gauge load cell platform.	24-bit differential ADC (HX711 or equivalent)	**MANDATORY**
FR-SENS-02	On-node digital filtering to remove floor vibration and electrical/ambient noise before weight derivation.	Median + moving average, or Kalman filter	**MANDATORY**
FR-SENS-03	Real-time item quantity derivation from calibrated baseline mass.	Qty = (Current Weight − Tare Weight) ÷ Unit Weight	**MANDATORY**
FR-SENS-04	Item identification and SKU assignment per smart bin; tags/codes map to item ID, unit weight, and reorder threshold.	SPI RFID reader (MFRC522 / PN532) or QR code	**HIGH**
FR-SENS-05	Continuous environmental monitoring (temperature and relative humidity).	I²C digital sensor (DHT22 / SHT30)	**MANDATORY**

1.2 Local Edge Processing & Fault Tolerance

Edge nodes must make local decisions, manage alarms, and avoid data loss during power or network outages.

ID	Description	Target Mechanism	Priority
FR-EDGE-01	Movement event logging: detect additions/removals, timestamp, quantity delta, and user ID if an RFID badge is scanned.	ESP32 interrupt logic	**MANDATORY**
FR-EDGE-02	Local anomaly detection (unauthorized access, off-hours or unusual bulk removal, over-temperature, high humidity) with local alarm and immediate alert packet.	Onboard buzzer and status LED	**MANDATORY**
FR-EDGE-03	Offline-first telemetry and transaction queuing during Wi-Fi or internet outages.	Non-volatile SPI flash / EEPROM queue	**MANDATORY**

ID	Description	Target Mechanism	Priority
FR-EDGE-04	Automatic synchronization and flushing of buffered data on network restoration.	Sequential MQTT re-sync engine	**MANDATORY**

1.3 Cloud Platform, Connectivity & Analytics

The backend provides secure communication, storage, analytics, dashboards, multilingual UI, and alert channels.

ID	Description	Target Mechanism	Priority
FR-CLD-01	Secure encrypted dispatch of telemetry, heartbeats, and event data using a topic-based architecture.	MQTT over TLS with client authentication	**MANDATORY**
FR-CLD-02	Trilingual user interface for diverse shop-floor operators.	React UI (English, Sinhala, Tamil)	**MANDATORY**
FR-CLD-03	Demand forecasting: learn consumption velocity, predict stock run-out dates, and recommend reorder quantities from lead times and safety stock levels.	Python (Holt-Winters / ML model)	**MANDATORY**
FR-CLD-04	Multi-channel instant alerts for low stock, power outage, and theft/anomaly events.	Email, SMS, and WhatsApp gateway API	**HIGH**
FR-CLD-05	Dashboard showing real-time inventory counts, movement history, active alerts, and environmental conditions across multiple sites and bins.	Web dashboard (React UI)	**MANDATORY**
FR-CLD-06	Simple network provisioning of new devices.	Wi-Fi access point / BLE portal	**HIGH**
FR-CLD-07	Secure over-the-air firmware updates triggered by cloud commands.	Cloud-triggered OTA update	**HIGH**

2. Performance & KPI Specifications

≤ ±1.0%	≤ 2.0 s	≥ 8 Hrs	< 15%
SCALE PRECISION	ALERT LATENCY	BATTERY BACKUP	FORECAST ERROR

ID	Performance Specification	Verification Method
PR-PERF-01	Measurement accuracy: within ±1% of full-scale or actual load after calibration.	Test bench calibration with certified masses
PR-PERF-02	Alert latency: event-driven MQTT messages (stock removal, anomaly) published to the broker within ≤ 2.0 s under standard Wi-Fi.	Event-to-broker timestamp comparison
PR-PERF-03	Sampling frequency: 10–80 Hz dynamic switching during active movement; low-power polling when idle.	Oscilloscope and logic analyzer bus log
PR-PERF-04	Offline queue buffer: minimum 10,000 events stored without data corruption or dropped packets.	Simulated 48-hour network drop test
PR-PERF-05	System uptime reliability: 99.5% continuous operation under standard grid power.	14-day continuous endurance burn-in

ID	Performance Specification	Verification Method
PR-PERF-06	Forecast quality: average run-out prediction error under 15%.	Back-test against historical consumption data

3. Cost, Power & Environmental Constraints

3.1 Cost Constraints

ID	Requirement
COST-01	**Prototype BOM Cost:** Target Bill of Materials cost per node shall remain within LKR 8,000 – 12,000.
COST-02	**Target Selling Price:** Commercial sale price per node shall be LKR 28,000 – 45,000, scaled by load capacity (5 kg / 20 kg / 50 kg).
COST-03	**Operational Server Cost:** Cloud backend (MQTT broker, PostgreSQL/TimescaleDB, API) shall be containerized with Docker for deployment on low-cost VPS hosting.

3.2 Power Architecture

ID	Requirement
PWR-01	**Primary Input:** 5V DC via USB-C (or micro-USB) with surge, overvoltage, and reverse-polarity protection.
PWR-02	**Battery Backup:** Integrated Li-ion charging and management circuit with seamless power-path switching.
PWR-03	**Outage Runtime:** On mains failure, the node shall switch to battery automatically, send a power-outage alert over MQTT, and operate for at least 8 hours.
PWR-04	**Consumption Profile:** The node shall meet these limits: • Active sensing & Wi-Fi transmission: peak ≤ 2.5 W (500 mA @ 5V) • Idle monitoring: average ≤ 0.5 W (100 mA @ 5V) • Deep sleep / power-saving: ≤ 10 mA between sample intervals

3.3 Environmental Tolerance

ID	Requirement
ENV-01	**Operating Temperature:** Continuous operation in indoor ambient temperatures of 0 °C to 50 °C (tropical climates, commercial kitchens, warehouses).
ENV-02	**Relative Humidity:** Tolerance of non-condensing humidity from 10% to 90% RH.
ENV-03	**Enclosure Protection:** 3D-printed/molded housing shall be splash-resistant and dust-protected (target IP54 equivalent) for cafés, pharmacies, and factory floors.
ENV-04	**Vibration Isolation:** Load cell platform mountings shall include mechanical dampening against floor vibration from machinery or nearby activity.